IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

ELECTROMAGNETICS

Tuesday, 4 May 10:00 am

Time allowed: hours

There are THREE questions on this paper.

Answer questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : R.R.A. Syms

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ELECTROMAGNETISM

1. a) A two-dimensional static electric field has a potential in the form $\varphi(x,y) = x+y$

i) Find the expression for the electric field vector $\mathbf{E}(x,y)$ [3]

ii) Find the expression for the field lines of $\mathbf{E}(x,y)$ [4]

iii) Sketch the field lines of $\mathbf{E}(x,y)$ [4]

iv) Find the volume charge density $\rho(x,y)$ [4]

b) Two *LC* transmission lines are connected together forming a junction. The lines have the same inductances per unit length equal to L. Line 1 has a capacitance per unit length of C_1, and line 2 has a capacitance per unit length of $C_2 = 4C_1$. Find the wave reflection and transmission coefficients when a wave is incident upon the junction from line 1. [7]

c) The dispersion relation for modes in a parallel-plate waveguide has the form

$$\omega = c\sqrt{k_z^2 + \left(\frac{n\pi}{d}\right)^2}$$

where ω is the angular frequency, c is the light velocity, k_z is the propagation constant, d is the waveguide height, and n is an integer. Find an expression for the propagation velocity of the modes in the limit $k_z \to \infty$ and give physical interpretation of the result. [10]

d) The components of the field of a short dipole antenna can be written in the form

$$H_\varphi = \frac{I_0 d}{4\pi}\frac{e^{-jkr}}{r}\left(jk + \frac{1}{r}\right)\sin\theta$$

$$E_r = -j\frac{I_0 d}{2\pi\varepsilon_0\omega}\frac{e^{-jkr}}{r^2}\left(jk + \frac{1}{r}\right)\cos\theta$$

$$E_\theta = -j\frac{I_0 d}{4\pi\varepsilon_0\omega}\frac{e^{-jkr}}{r}\left(-k^2 + \frac{jk}{r} + \frac{1}{r^2}\right)\sin\theta$$

Derive two sets of approximate expressions for these field components that are valid in the following two regimes: $kr \gg 1$ (far-field region) and $kr \ll 1$ (near-field region). Compare and contrast the two sets to each other. [8]

2. a) A plane linearly polarised TEM wave is incident normally upon a flat boundary of a material. The electromagnetic wave cannot penetrate this material, and the boundary condition for it demands that the tangential component of the *magnetic* field vector **H** vanishes at the boundary. Assuming a trial electric field vector **E**, derive an expression for the wave reflection coefficient, and hence construct expressions for the total vectors **E** and **H**. [15]

b) The two-port device shown in Fig. 2.1 is to be connected to 50-Ω transmission lines. Find the scattering matrix of the device. What function does the device perform? [15]

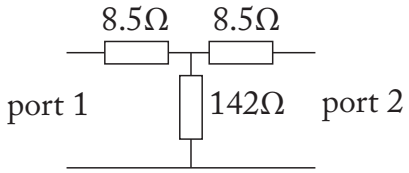
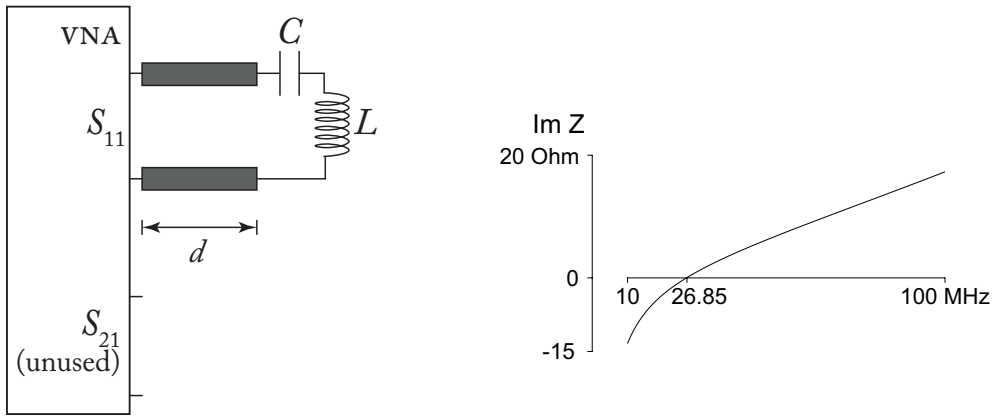


Figure 2.1

3. A student is measuring S_{11} between 10 MHz and 100 MHz in the configuration shown in Fig. 3.1(a). In this configuration, an *LC* resonator is connected to the output port of a 50-Ω network analyser via a matched coaxial cable. The cable length is $d = 15$ cm, and the dielectric between its conductors has a relative permittivity equal to 1. The student knows the inductance L to be equal to 10 nH.

The student then converts the measured S_{11} into input impedance Z. The student sees that the real part of the impedance is equal to zero across the entire frequency range. The result for the imaginary part of the impedance is shown in Fig. 3.1(b).



(a) (b)

Figure 3.1

a) Find the values of $|S_{11}|$ across the entire frequency range. [2]

b) Derive the expression that the student used to calculate the impedance Z from the measured values of S_{11}. [8]

c) Find the value of the capacitance C. [8]

d) The student then decides to replace the cable with another one with the same properties but with a length of 10 cm. Using the same axes as in Fig. 3.1(b), sketch qualitatively the imaginary part of the impedance that will be obtained. Explain your reasoning. [12]